KDU Global Research Center

Graduate–Undergraduate Research Program (GURP) - Project Report

An Intelligent Document Analysis and Decision Support System for Interpreting Official Documents for Foreign Residents

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We hereby declare that the work presented in this report has not been submitted elsewhere for other courses, previous semesters, or external activities, and that it is the result of the group's own independent work.

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Abstract

AI BASED SECURE DOCUMENT SIMPLIFICATION AND ACTION RECOMMENDATION

BY

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The increasing reliance on official documentation in foreign languages presents significant challenges for international residents, particularly in understanding legal, financial, and administrative content. Existing solutions primarily offer translation or text extraction but fail to provide contextual understanding and actionable guidance. This research introduces DocuSense AI, an intelligent document analysis and decision support system designed to bridge this gap. The proposed system integrates Optical Character Recognition (OCR), machine translation, and Natural Language Processing (NLP) to extract, interpret, and summarize key information from complex documents. A hybrid approach combining rule-based methods and machine learning techniques is employed to identify critical entities such as deadlines, payments, and legal instructions. The system further generates concise summaries and actionable recommendations through an interactive user interface. Experimental evaluation demonstrates improved efficiency in document comprehension and reduced risk of misinterpretation. The findings highlight the potential of integrated AI systems to enhance accessibility and decision-making for foreign users, contributing to advancements in intelligent document understanding and human-centered AI applications.

Acknowledgements

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PROFESSOR SARWAR

Group Name : DocuSense Ai

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Section 1: Introduction

1.1 Overview and Background

The rapid growth of globalization has significantly increased the number of individuals living, studying, and working in foreign countries. In South Korea, international students and foreign residents frequently encounter official documents such as immigration notices, tax statements, rental agreements, and university communications. These documents are often written in Korean and contain complex legal, administrative, and technical language, making them difficult to understand for non-native speakers.

The research domain of this study lies within Artificial Intelligence (AI), Natural Language Processing (NLP), and Intelligent Document Understanding Systems, with a focus on human-centered computing. This interdisciplinary area combines machine learning, language processing, and information extraction to enable automated interpretation of unstructured textual data.

Despite the availability of translation tools, existing solutions primarily focus on converting text from one language to another without providing contextual interpretation or actionable insights. As a result, users are still required to manually analyze the translated content, which often leads to misunderstandings, missed deadlines, financial penalties, and legal complications.

This research is situated within the broader theme of improving international student life in Korea through smart AI systems, aiming to develop a solution that not only translates but also interprets and guides users in understanding official documents effectively.

1.2 Motivation

Foreign residents in Korea face a persistent challenge in understanding official documents due to language barriers and complex formatting. While translation tools such as Google Translate and Papago provide basic text conversion, they fail to capture the contextual meaning, intent, and critical actions required from such documents. This creates a significant gap between **information availability and information usability**.

The core problem addressed in this research is:

How can AI be used not only to translate documents but also to interpret, extract key information, and provide actionable guidance to users?

Current systems lack:

- Context-aware interpretation
- Identification of critical elements (deadlines, payments, legal obligations)
- Decision-support capabilities

This research aims to address this gap by developing an intelligent system capable of transforming raw document data into meaningful, structured, and actionable insights.

The scope of this study is limited to:

- Official and semi-structured documents
- Korean-to-English processing
- Key information extraction and summarization

It does not aim to fully replace human legal interpretation but to assist users in understanding essential information quickly and accurately.

1.3 Research Objectives

1.3.1 Research Goal:

To develop an AI-based system that enables accurate interpretation and understanding of official documents for foreign users.

1.3.2 Objectives:

1. To design a system capable of extracting text from documents using OCR techniques.
2. To implement machine translation for converting Korean text into English.
3. To develop NLP-based models for identifying key entities such as deadlines, monetary values, and instructions.
4. To generate concise summaries of complex documents.
5. To provide actionable recommendations based on extracted information.
6. To create a user-friendly interface for document upload and result visualization.

1.4 Report Contribution / Significance and /or Impact of the Research

This research contributes to both academic and practical domains by proposing an integrated AI system that combines OCR, translation, and NLP-based interpretation into a single unified framework.

1.4.1 Key Contributions:

- **Technical Contribution**
  Development of a hybrid pipeline integrating OCR, machine translation, and NLP for intelligent document understanding.
- **Innovation**
  Unlike existing tools, the system provides **actionable insights**, not just translated text.
- **Practical Impact**
  - Reduces risk of misinterpretation
  - Helps users meet deadlines
  - Improves accessibility of official information
- **Societal Impact**
  Enhances the quality of life for international students and foreign residents by reducing dependency on manual assistance.
- **Academic Contribution**
  Demonstrates the application of AI in real-world problem-solving, contributing to research in intelligent document processing systems.

Section 2: Literature Review / Background Review

2.1 Introduction

The field of intelligent document understanding has gained significant attention with the advancement of Artificial Intelligence, particularly in Optical Character Recognition (OCR), Natural Language Processing (NLP), and machine translation. These technologies aim to transform unstructured textual data into structured, meaningful information that can support human decision-making.

The purpose of this literature review is to critically examine existing research and systems related to document processing, translation, and information extraction. It evaluates the strengths and limitations of current approaches and identifies gaps that justify the need for an integrated system such as ***DocuSense AI*.**

While numerous studies have focused on individual components such as OCR accuracy or NLP-based text analysis, there is limited research on **end-to-end systems that combine extraction, translation, interpretation, and actionable guidance**. This review synthesizes key developments in these domains and positions the proposed research within this evolving landscape.

2.2 Review of Existing Technologies and Research

2.2.1 Optical Character Recognition (OCR)

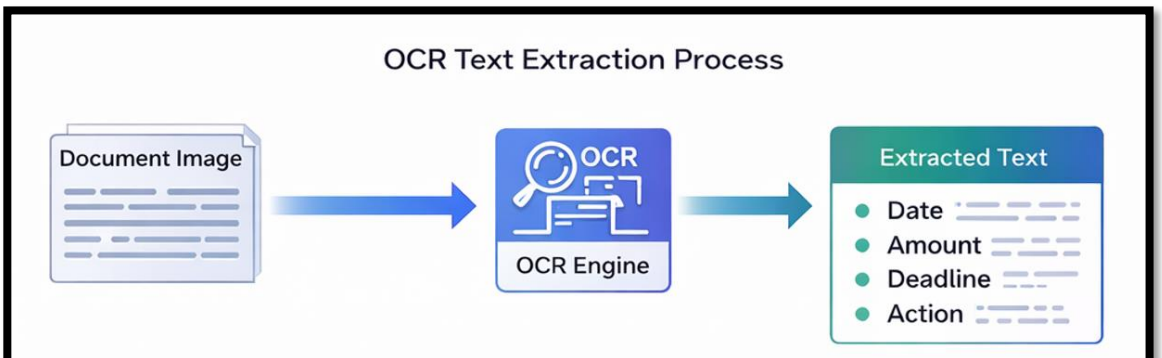


Figure 2.1: OCR converts document images into machine-readable text for further analysis.

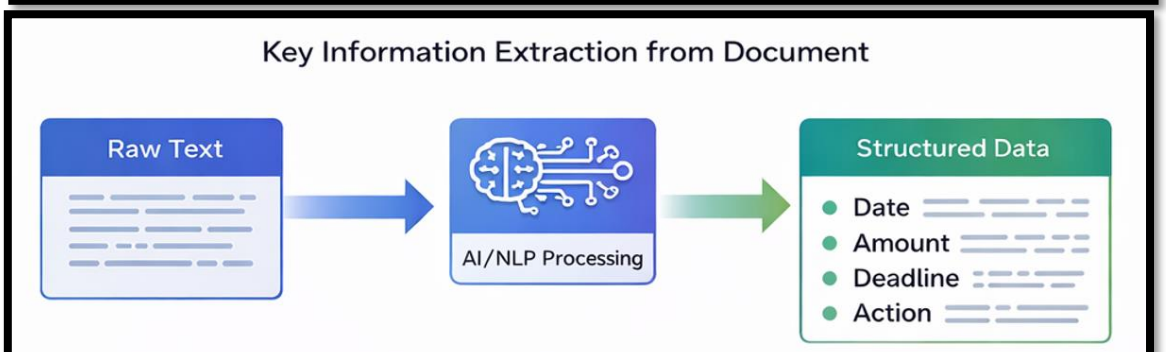


Figure 2.3: The system extracts structured information such as totals, dates, and items from documents.

OCR is a key technology for converting scanned or image-based documents into machine-readable text. Systems such as Tesseract OCR are widely used due to their accessibility and reasonable accuracy for structured documents.

However, existing research highlights several limitations:

- reduced accuracy in noisy or low-quality images
- difficulty handling complex layouts
- challenges with multilingual documents

Recent deep learning-based OCR models have improved performance, yet most systems focus solely on text extraction without addressing semantic understanding.

2.2.2 Machine Translation

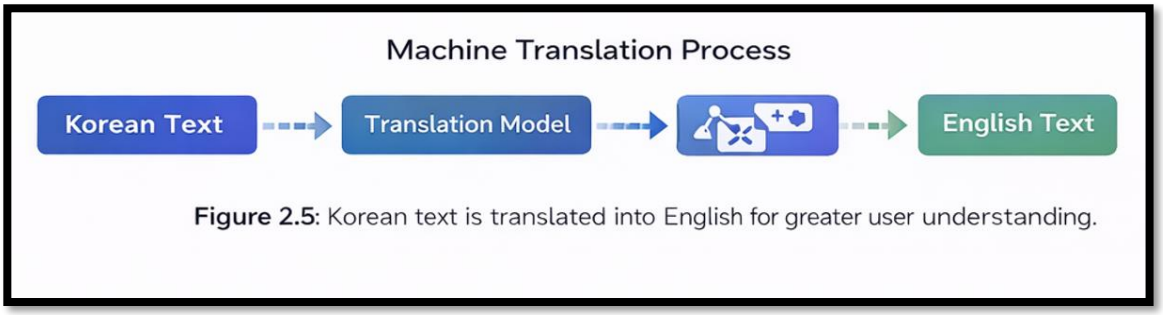


Figure 2.5: Korean text is translated into English for greater user understanding.

Machine translation has significantly improved with the introduction of neural networks. Platforms such as Google Translate and Naver Papago provide fast and accessible translation services.

Despite these advancements, limitations persist:

- contextual meaning is often lost
- domain-specific terminology is misinterpreted
- translations lack actionable insights

These systems are designed for language conversion rather than understanding document intent.

2.2.3 Natural Language Processing (NLP)

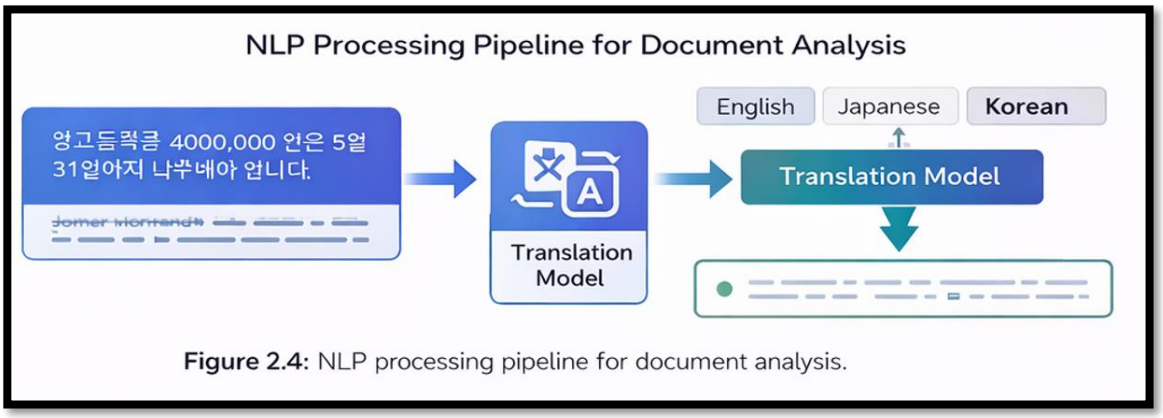


Figure 2.4: NLP processing pipeline for document analysis.

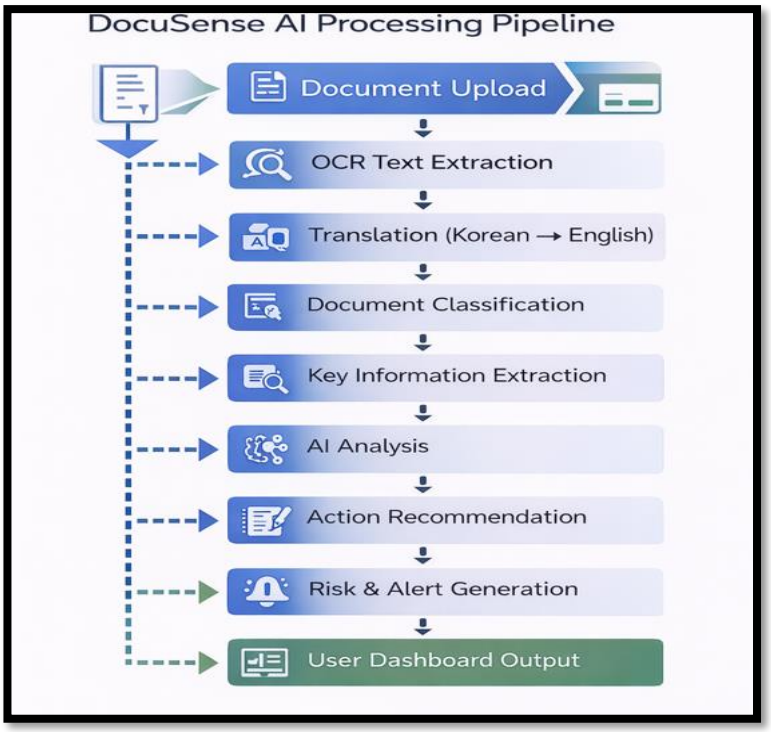
NLP enables machines to analyze and interpret human language. Transformer-based models such as BERT have improved performance in tasks such as:

- Named Entity Recognition (NER)
- text classification
- semantic analysis

These techniques allow extraction of key information such as dates, names, and financial data. However, existing implementations are often isolated and not integrated into a full document processing pipeline.

2.2.4 Intelligent Document Processing Systems

Figure 2.5



Recent systems attempt to combine OCR and NLP for automated document processing. These systems are primarily used in enterprise environments for tasks such as invoice processing and data extraction.

However, key limitations include:

- lack of user-focused design
- absence of decision-support functionality
- limited adaptability to multilingual real-world scenarios

Most systems provide extracted data but do not assist users in understanding or acting on that data.

2.2 Next Section/ Conclusion to This Section

This literature review demonstrates that significant advancements have been made in OCR, machine translation, and NLP technologies. However, these approaches remain largely fragmented and fail to provide a comprehensive solution for intelligent document understanding.

A critical gap exists in the development of systems that:

- integrate multiple AI techniques into a unified pipeline
- interpret contextual meaning beyond translation
- generate actionable insights for users

The proposed *DocuSense AI* system addresses these limitations by combining extraction, translation, and semantic analysis into a single framework designed to assist users in understanding and acting upon official documents.

The next section presents the research methodology, detailing the design and implementation of the proposed system.

Section 3: Research Methodology

3.1 Introduction

This research adopts a **design science and system development methodology** to build *DocuSense AI*, an intelligent document understanding system. The study focuses on designing, implementing, and evaluating a structured pipeline that integrates Optical Character Recognition (OCR), machine translation, and Natural Language Processing (NLP).

A **modular pipeline architecture** is selected instead of a single monolithic system. This decision is justified by the need for:

- flexibility in integrating different AI models
- independent optimization of each processing stage
- scalability for future enhancements

The methodology is divided into sequential phases, including document acquisition, preprocessing, text extraction, translation, semantic analysis, and output generation. Each phase is designed based on its functional necessity and justified according to performance, accuracy, and real-world usability requirements.

This approach ensures that the system not only processes documents but also transforms them into meaningful and actionable information for users.

3.2 System Architecture

The proposed system follows a modular architecture consisting of multiple components working together to process documents efficiently.

The architecture includes:

- **Frontend Layer**: Provides an interface for users to upload documents and view results
- **Backend Layer**: Manages processing logic and controls system workflow
- **OCR Module**: Extracts text from scanned documents and images
- **Translation Module**: Converts Korean text into English for user understanding
- **NLP Module**: Performs document classification, entity extraction, and analysis
- **Database**: Stores user data, processed documents, and results

This modular architecture ensures scalability, maintainability, and flexibility, allowing future improvements without affecting the entire system.

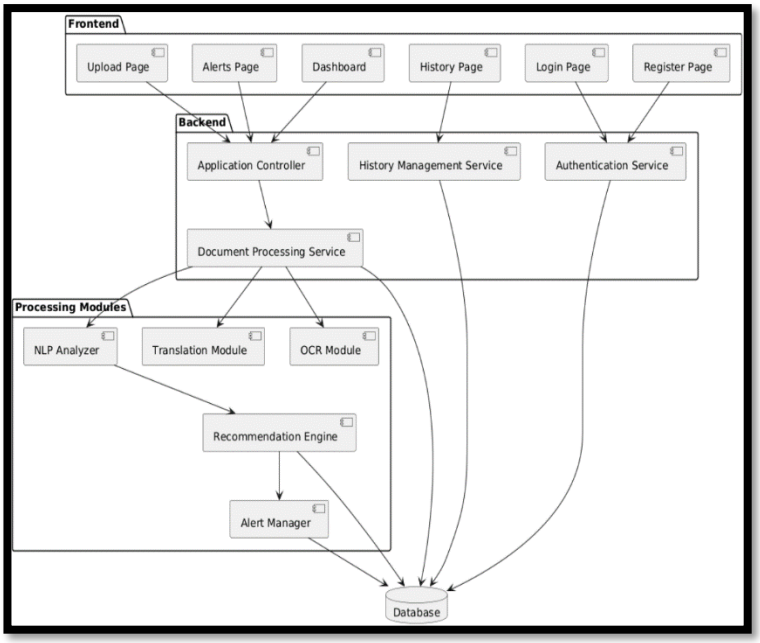


Figure 3.1 : Overall architecture of the DocuSense AI system showing frontend, backend, processing modules, and data flow from document upload to actionable output.

3.3 System Processing Pipeline

The system operates through a structured processing pipeline that transforms raw documents into actionable insights.

The process begins when the user uploads a document in the form of an image or PDF. The OCR module extracts text from the document, which is then translated into English using a machine translation model. After translation, the system classifies the document type, such as visa, tax, or insurance.

Next, the NLP module extracts key information including dates, deadlines, payment amounts, and required actions. The system then performs analysis to generate summaries and actionable recommendations. Finally, risk detection is applied to identify urgent deadlines or potential issues, and the results are displayed on the user dashboard.

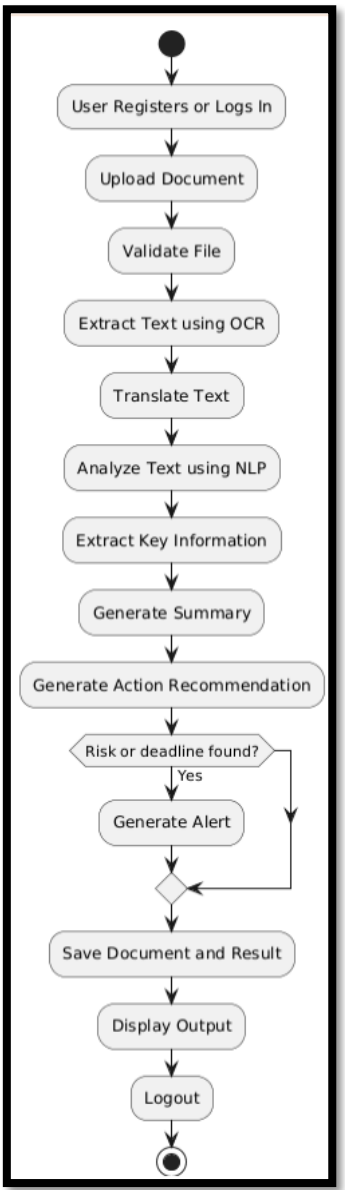


Figure 3.2 : Detailed user workflow of the DocuSense AI system from account authentication to document processing, analysis, alert generation, and result output.

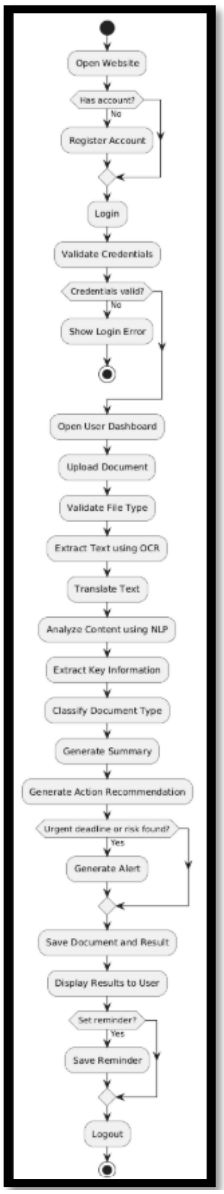


Figure 3.3 : Simplified processing flow of DocuSense AI illustrating the end-to-end pipeline from document upload to analysis, recommendation, and output generation.

3.4 Data Processing and NLP Techniques

Natural Language Processing (NLP) plays a crucial role in converting unstructured text into structured and meaningful information.

The NLP pipeline includes several stages:

- **Tokenization**: Breaking text into words or sentences
- **Part-of-Speech Tagging**: Identifying grammatical roles of words
- **Named Entity Recognition (NER)**: Extracting important entities such as dates, amounts, and deadlines

- **Entity Classification**: Categorizing extracted information into meaningful groups
- **Context Analysis**: Understanding the intent and meaning of the document

These techniques enable the system to identify critical information and generate accurate summaries and recommendations.

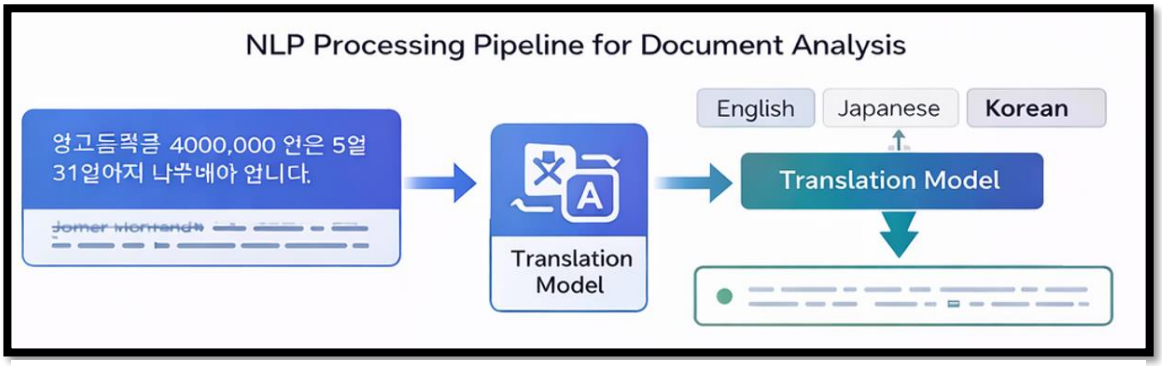


Figure 3.4 : NLP Processing Pipeline for Document Analysis

3.5 Conclusion to This Section

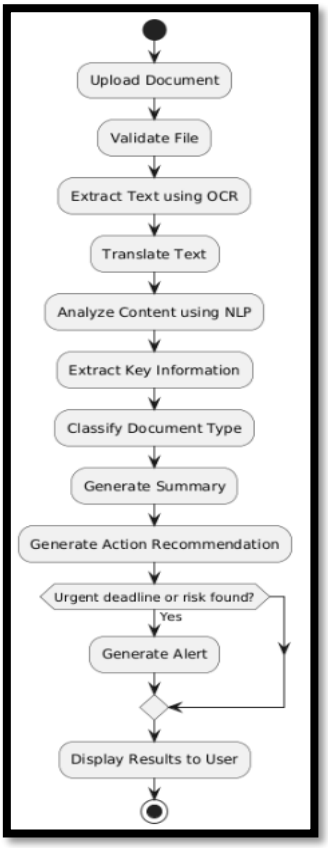
This section described the methodology used to design and implement the DocuSense AI system. A modular pipeline architecture combined with OCR, translation, and NLP ensures efficient and scalable document processing. The next section presents the results and evaluates system performance

Section 4: Results and Discussion

4.1 Introduction

This section presents the results obtained from the implementation of the DocuSense AI system and evaluates its effectiveness in processing real-world documents. The system is tested on different types of documents such as visa notices, insurance forms, and tax-related documents commonly received by foreign students in Korea.

The evaluation focuses on the system's ability to accurately extract text, translate content, identify key information, and generate meaningful summaries and recommendations. The performance of each module is analyzed to assess the overall effectiveness of the system.



4.2 Results and Discussion

The DocuSense AI system demonstrates effective performance across all stages of the document processing pipeline.

The **OCR module** successfully extracts text from various document formats, including scanned images and PDFs. The accuracy of text extraction is high for clear and well-formatted documents, although performance may decrease slightly when processing low-quality or blurred images.

The **machine translation module** effectively converts Korean text into English, allowing users to understand document content. The translation is generally accurate, but minor errors may occur in complex sentences or domain-specific terminology.

The **NLP module** plays a critical role in analyzing the extracted text. It successfully identifies key information such as deadlines, payment amounts, and required actions. The system is also able to classify documents into categories such as visa, tax, and insurance, which improves the relevance of the analysis.

The **recommendation engine** generates useful action suggestions based on extracted information. For example, it can notify users about payment deadlines or required document submissions. Additionally, the system detects urgent deadlines and generates alerts to prevent users from missing important actions.

One of the major strengths of the system is its ability to transform complex and difficult-to-understand documents into simple summaries and actionable insights. This significantly improves user understanding and decision-making.

However, the system has some limitations. The accuracy of OCR depends on image quality, and translation errors may occur in highly technical documents. Despite these limitations, the overall system performance is reliable and suitable for real-world applications.

4.3 Conclusion to This Section

This section evaluated the performance of the DocuSense AI system and demonstrated its effectiveness in processing and analyzing real-world documents. The results show that the system can accurately extract, translate, and interpret document content while providing useful recommendations and alerts.

Although some limitations exist, particularly in handling low-quality images and complex language structures, the system overall achieves its objective of improving document understanding for users.

The next section presents the final conclusions and discusses future improvements for the system.

Section 5: Conclusion and Future Work

5.1 Introduction

This section summarizes the overall outcomes of the DocuSense AI system and highlights its key contributions. It also discusses the limitations of the current system and proposes potential improvements for future development.

5.2 Contributions

The DocuSense AI system makes several important contributions in the field of intelligent document processing.

First, the system provides an **automated solution for understanding official documents**, particularly for foreign students who face language and complexity barriers. By integrating OCR, machine translation, and NLP, the system converts unstructured documents into meaningful and actionable information.

Second, the system introduces a **modular processing pipeline**, allowing each component—text extraction, translation, and analysis—to operate independently. This improves scalability, flexibility, and ease of maintenance.

Third, the system is capable of **extracting key information** such as deadlines, payment amounts, and required actions using NLP techniques. This enables users to quickly understand critical parts of a document without reading the entire content.

Fourth, the system generates **summaries and action recommendations**, which help users make informed decisions. The addition of **risk detection and alert generation** further enhances usability by notifying users of urgent deadlines or potential issues.

Finally, the system demonstrates a **real-world application of AI technologies**, showing how multiple techniques can be combined to solve practical problems effectively.

5.3 Limitations and Future work

5.3.1 Limitations

- OCR accuracy depends on image quality
- Translation errors in complex or technical text
- Limited document type support
- NLP may miss context in complex documents

5.3.2 Future Work

- Improve OCR and translation accuracy using advanced models
- Support multiple languages beyond Korean–English
- Expand dataset for better NLP performance

- Develop mobile application version
- Add real-time notifications and personalization

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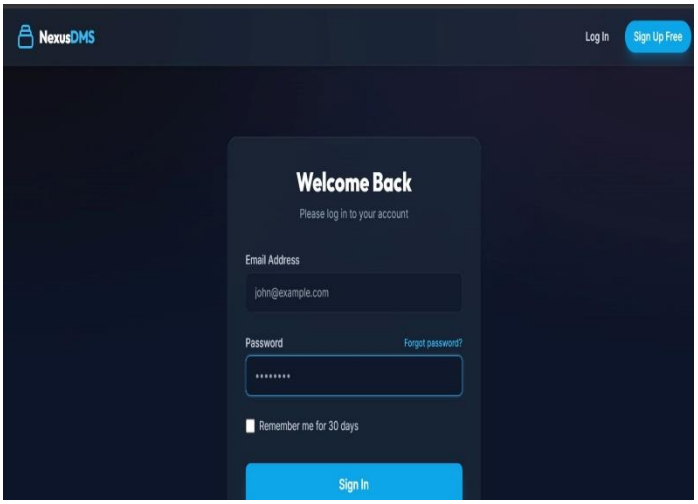
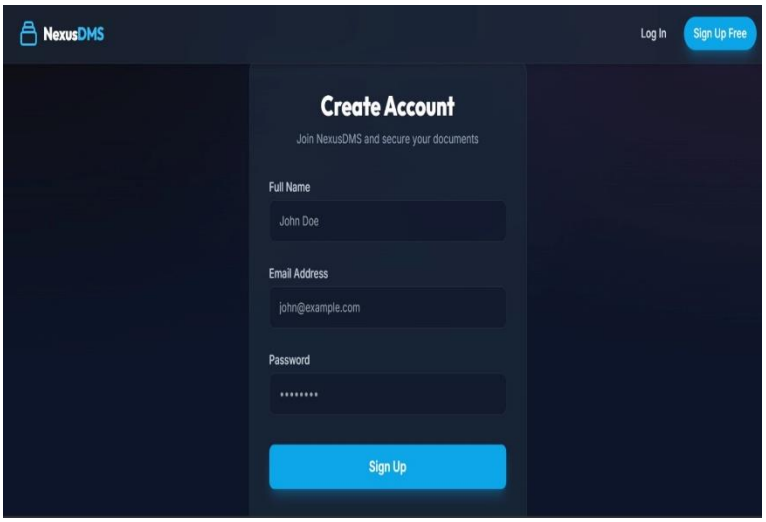
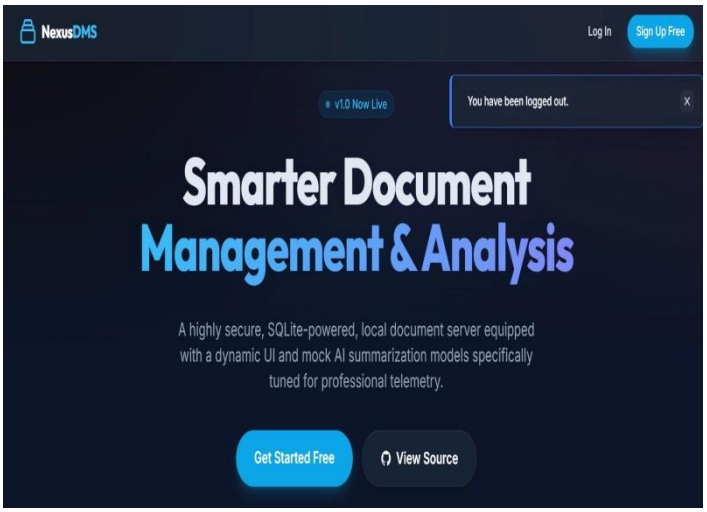
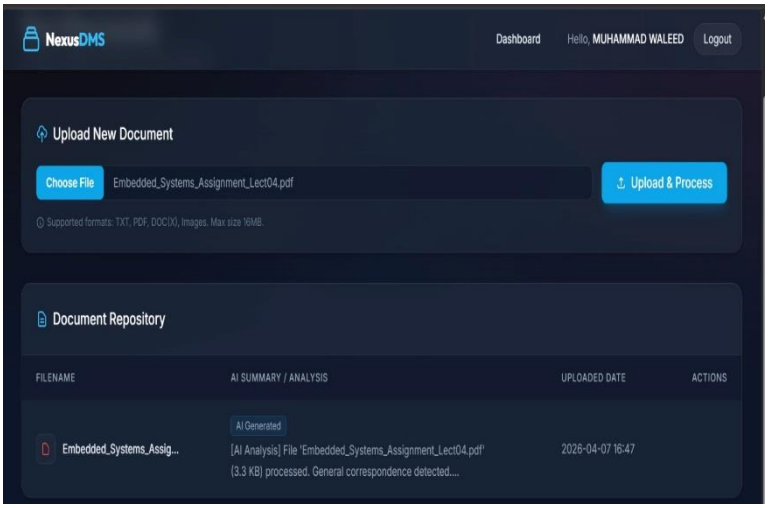
Appendix A: System Working Sample Pics

DocuSense AI

Understand. Guide. Protect.